

Assignment 1: Detailed Solutions

Statistical Inference 2026–27

These notes give the arguments behind the conclusions, including the assumptions needed for conditional expectations and the Cramér–Rao bound. Numerical results are one reproducible realization of the requested experiment; another random seed will produce different counts and estimates.

Problem 1: Completeness and minimal sufficiency

Correction to the statement. The result as written is false: *complete* must be replaced by *complete and sufficient*. Completeness alone does not imply sufficiency. For example, let $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$, and let $T(X) = 0$. If $\mathbb{E}_\theta h(T) = 0$ for every θ , then $h(0) = 0$, so T is complete. However, T is not sufficient, since conditioning on a constant leaves the distribution of X dependent on θ . A minimal sufficient statistic exists: X itself. Thus this example contradicts the statement in the assignment. Indeed, the likelihood ratio at observations x, y is $\exp\{\theta(x - y) - (x^2 - y^2)/2\}$, independent of θ precisely when $x = y$, verifying minimal sufficiency by the likelihood-ratio criterion.

The corrected result is:

If a minimal sufficient statistic exists, every complete sufficient statistic is minimal sufficient.

Definitions and measurable conventions

A statistic T is complete if, for every measurable h such that $\mathbb{E}_\theta |h(T)| < \infty$ for all θ ,

$$[\mathbb{E}_\theta h(T) = 0 \text{ for every } \theta] \implies \mathbb{P}_\theta\{h(T) = 0\} = 1 \text{ for every } \theta.$$

It is *boundedly complete* if this implication is required only for bounded measurable h . Thus completeness implies bounded completeness. Neither definition says that T itself must be bounded or integrable.

A sufficient statistic S is minimal sufficient if, for every sufficient statistic R , there is a measurable function f_R , independent of θ , such that $S = f_R(R)$ almost surely under every $\mathbb{P}_\theta$. We use the customary measurable setting for statistical models: real-valued or finite-dimensional statistics, with parameter-independent versions of conditional distributions provided by sufficiency. “Almost surely for every θ ” means that the same measurable equality has probability one under each member of the family.

Proof without any moment assumption on the statistic

Proof. First suppose that T is real-valued. Let S be minimal sufficient and T complete and sufficient. Minimal sufficiency gives a parameter-independent measurable function f such that $S = f(T)$ almost surely for every θ .

Define the bounded, one-to-one function

$$b(t) = \arctan t, \quad |b(t)| < \frac{\pi}{2}.$$

Because S is sufficient, there is a measurable function H , independent of θ , such that

$$H(S) = \mathbb{E}_\theta[b(T) \mid S] \quad \text{for every } \theta.$$

It can be chosen with $|H| \leq \pi/2$: use a common conditional distribution of the sample given S to integrate $b(T)$, and define $H = 0$ where necessary. In particular, all expectations below exist. The tower property gives

$$\mathbb{E}_\theta H(S) = \mathbb{E}_\theta b(T).$$

Set $d(t) = b(t) - H(f(t))$. This is a parameter-independent bounded measurable function, and

$$\mathbb{E}_\theta d(T) = \mathbb{E}_\theta \{b(T) - H(S)\} = 0 \quad \text{for every } \theta.$$

Completeness of T implies

$$\arctan T = H(S) \quad \text{almost surely for every } \theta.$$

Define the measurable function

$$k(s) = \begin{cases} \tan H(s), & |H(s)| < \pi/2, \\ 0, & |H(s)| \geq \pi/2. \end{cases}$$

Since $H(S) = \arctan T \in (-\pi/2, \pi/2)$ almost surely, we have $T = k(S)$ almost surely for every θ .

Now let R be any sufficient statistic. Since S is minimal sufficient, $S = f_R(R)$ almost surely. Hence

$$T = k(S) = k(f_R(R)) \quad \text{almost surely for every } \theta.$$

This is precisely the defining property of minimal sufficiency for T .

For a finite-dimensional statistic $T = (T_1, \dots, T_d)$, apply the same argument to $\arctan T_j$, one coordinate at a time, while using completeness of the *whole* statistic T . The finite collection of equalities yields $T = k(S)$ as a vector. $\square$

What role did boundedness play? We bounded a *transformation*, not the statistic. This guarantees integrability and lets us invoke completeness without assumptions about $\mathbb{E}_\theta|T|$ or $\mathbb{E}_\theta T^2$. In fact the proof uses only bounded completeness. If T is integrable under every parameter value, $b(t) = t$ also works with ordinary completeness. A proof based on total variance would instead require square integrability of the random variables to which that identity is applied. No Rao–Blackwell or Lehmann–Scheffé theorem is needed for the argument above.

Problem 2: MLE, the Cramér–Rao bound, and UMVUE

The CRLB and its equality condition

Let $f_\theta(x)$ denote the joint density or mass function of the sample with respect to a parameter-independent dominating measure μ . For this discussion, assume the parameter space is an open interval, the support is fixed, and f_θ is positive on that support and differentiable in θ . Assume that differentiating under the integral is valid both for $\int f_\theta d\mu$ and for the expectations of the estimators

used below. Local integrable bounds on the corresponding derivatives are one way to justify these interchanges. Also suppose

$$0 < I_n(\theta) := \mathbb{E}_\theta[U(\theta; X)^2] < \infty.$$

These are sufficient regularity assumptions for the following calculations; they are not claims about every parametric model.

The score has mean zero because

$$\mathbb{E}_\theta U(\theta; X) = \int \frac{\partial_\theta f_\theta(x)}{f_\theta(x)} f_\theta(x) d\mu(x) = \frac{d}{d\theta} \int f_\theta(x) d\mu(x) = 0.$$

Let W be an unbiased estimator of θ with finite variance. Then

$$\begin{aligned} 1 &= \frac{d}{d\theta} \mathbb{E}_\theta W = \int W(x) \partial_\theta f_\theta(x) d\mu(x) \\ &= \mathbb{E}_\theta[WU(\theta; X)] = \mathbb{E}_\theta[(W - \theta)U(\theta; X)]. \end{aligned}$$

Applying Cauchy–Schwarz gives

$$1 \leq \mathbb{E}_\theta[(W - \theta)^2] \mathbb{E}_\theta[U^2] = \text{Var}_\theta(W) I_n(\theta),$$

so

$$\boxed{\text{Var}_\theta(W) \geq I_n(\theta)^{-1}.}$$

Equality in Cauchy–Schwarz holds exactly when the two random variables are proportional almost surely. If $U = a(\theta)(W - \theta)$, the covariance identity above gives $1 = a(\theta) \text{Var}_\theta(W)$. At equality,

$$\boxed{U(\theta; X) = I_n(\theta)(W - \theta) \quad \text{almost surely under } \mathbb{P}_\theta.}$$

This is an equality condition for a *specified unbiased estimator* at a specified parameter value. Attaining the CRLB throughout the parameter space requires this property for the same statistic W at every θ .

1. The logical flaw

The equality condition for the CRLB is a *special property* of an unbiased estimator that attains the bound. It does not hold automatically for an arbitrary MLE. The student assumes the existence of an unbiased estimator W satisfying

$$U(\theta; X) = a(\theta)\{W(X) - \theta\}$$

and then treats that assumption as a universal fact.

Even if such a representation exists, using it at a random value $\hat{\theta}_{\text{MLE}}$ requires an identity valid simultaneously over the parameter space on a common event of probability one. Also, the score equation holds for a differentiable *interior* maximum; an MLE on the boundary need not solve it.

The valid implication is conditional: if the regularity assumptions hold and an estimator W satisfies the CRLB equality representation, then an interior MLE solving the score equation must equal W , provided a is nonzero there. The score equation by itself does not establish the equality representation, unbiasedness, or efficiency.

It is also essential that W be a statistic, independent of the unknown parameter. For a fixed θ_0 the expression $\theta_0 + U(\theta_0; X)/I_n(\theta_0)$ can be written down, but a family of such expressions indexed by an unknown parameter does not automatically define one unbiased, efficient estimator on the whole parameter space.

2. Counterexample: estimating an exponential rate

Let $X_1, \dots, X_n$ be independent exponential random variables with rate $\theta > 0$, and suppose $n > 2$. Their density is

$$f_\theta(x) = \theta e^{-\theta x} \mathbf{1}\{x > 0\}.$$

Write $S = \sum_{i=1}^n X_i$. The likelihood and score are

$$L(\theta; x) = \theta^n e^{-\theta S}, \quad U(\theta; x) = \frac{n}{\theta} - S.$$

Solving $U = 0$ gives

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{n}{S}}.$$

The second derivative is $-n/\theta^2 < 0$, so this is the unique maximum.

Since $S \sim \text{Gamma}(n, \text{rate } \theta)$,

$$\mathbb{E}_\theta(S^{-1}) = \frac{\theta}{n-1}, \quad \mathbb{E}_\theta(S^{-2}) = \frac{\theta^2}{(n-1)(n-2)}.$$

For example, the first identity follows directly from

$$\mathbb{E}_\theta(S^{-1}) = \frac{\theta^n}{\Gamma(n)} \int_0^\infty s^{n-2} e^{-\theta s} ds = \frac{\theta^n \Gamma(n-1)}{\Gamma(n) \theta^{n-1}}.$$

Therefore

$$\mathbb{E}_\theta \hat{\theta}_{\text{MLE}} = \frac{n}{n-1} \theta \neq \theta.$$

The MLE is biased and hence cannot be the UMVUE of θ .

In fact,

$$\boxed{\tilde{\theta} = \frac{n-1}{S}}$$

is unbiased. The statistic S is sufficient by factorization and complete because this is a full one-parameter exponential family with natural parameter $-\theta$ ranging over the open interval $(-\infty, 0)$. Equivalently, completeness follows from uniqueness of the Laplace transform of $h(s)s^{n-1}$ when $\mathbb{E}_\theta h(S) = 0$ for all $\theta > 0$. The Lehmann–Scheffé theorem thus gives that $\tilde{\theta}$ is the UMVUE.

A direct completeness proof. Suppose h is measurable, $\mathbb{E}_\theta |h(S)| < \infty$ for every $\theta > 0$, and $\mathbb{E}_\theta h(S) = 0$ for every $\theta > 0$. The gamma density gives

$$\int_0^\infty h(s) s^{n-1} e^{-\theta s} ds = 0 \quad \text{for every } \theta > 0.$$

Fix $\theta_0 > 0$ and define the finite signed measure

$$d\nu(s) = h(s) s^{n-1} e^{-\theta_0 s} ds.$$

Its total variation is finite by the assumed integrability at θ_0 . Push this measure forward by $z = e^{-s}$ to a finite signed measure $\tilde{\nu}$ on $[0, 1]$, assigning zero mass to the endpoints if needed. For every integer $k \geq 0$,

$$\int_0^1 z^k d\tilde{\nu}(z) = \int_0^\infty h(s) s^{n-1} e^{-(\theta_0+k)s} ds = 0.$$

Thus every polynomial has zero integral. Polynomials are uniformly dense in the continuous functions on $[0, 1]$, and integration against a finite signed measure is continuous in the uniform norm. Consequently every continuous function has zero integral, which implies $\tilde{\nu} = 0$. The transformation is one-to-one with a measurable inverse on $(0, 1)$, so $\nu = 0$. Since $s^{n-1}e^{-\theta_0 s} > 0$ for $s > 0$, $h(s) = 0$ Lebesgue-almost everywhere. Every gamma density here is positive on $(0, \infty)$, giving $h(S) = 0$ almost surely for every θ .

Why completeness gives the UMVUE. For any other unbiased estimator V of θ with finite variance, sufficiency of S gives a parameter-independent conditional expectation $v(S) = \mathbb{E}_\theta[V \mid S]$. It is unbiased by the tower property. Both $v(S)$ and $(n-1)/S$ are integrable functions of S with the same expectation for every θ . Completeness therefore implies

$$v(S) = \frac{n-1}{S} \quad \text{almost surely for every } \theta.$$

The total-variance identity now yields

$$\text{Var}_\theta(V) = \text{Var}_\theta(v(S)) + \mathbb{E}_\theta \text{Var}_\theta(V \mid S) \geq \text{Var}_\theta\left(\frac{n-1}{S}\right).$$

This proves the UMVUE claim directly.

There is another useful distinction here. The sample Fisher information is $I_n(\theta) = n/\theta^2$, whereas

$$\text{Var}_\theta(\tilde{\theta}) = \frac{\theta^2}{n-2} > \frac{\theta^2}{n} = \frac{1}{I_n(\theta)}.$$

Thus an estimator can be UMVUE without attaining the CRLB.

For completeness, the displayed variance follows by subtraction:

$$\text{Var}_\theta\left(\frac{n-1}{S}\right) = (n-1)^2 \frac{\theta^2}{(n-1)(n-2)} - \theta^2 = \frac{\theta^2}{n-2}.$$

The Fisher information follows from $U = n/\theta - S$ and $\text{Var}_\theta(S) = n/\theta^2$. For this model an efficient unbiased estimator cannot exist throughout the parameter space: the equality representation would require

$$\frac{n}{\theta} - S = \frac{n}{\theta^2}(W - \theta), \quad \text{or} \quad W = 2\theta - \frac{\theta^2}{n}S,$$

whose right-hand side depends on θ . Because the sample distributions have common positive support, the almost-sure identities at two distinct parameters would force S to be a constant, contradicting its gamma law.

3. Sufficient conditions for an efficient, unbiased MLE

Here is one sufficient set of conditions. Assume:

1. The parameter space is an open interval; the model satisfies the usual CRLB regularity conditions, including differentiation under the integral, $\mathbb{E}_\theta U = 0$, and $0 < I_n(\theta) = \mathbb{E}_\theta U^2 < \infty$. For the UMVUE conclusion, the score covariance identity must hold for every finite-variance unbiased competitor. This is part of the regularity assumption for the class of estimators being compared; if it is imposed only on a smaller class, the minimum-variance conclusion is restricted to that class.

2. There is a statistic W , with finite second moment, and a deterministic nonzero function a such that, simultaneously for every θ ,

$$U(\theta; X) = a(\theta)\{W(X) - \theta\}$$

on a common event of probability one.

3. An MLE exists in the interior almost surely and satisfies its score equation.

Taking expectations of the representation gives

$$0 = a(\theta)\{\mathbb{E}_\theta W - \theta\}, \quad \text{so} \quad \mathbb{E}_\theta W = \theta.$$

Differentiating the unbiasedness identity under the integral yields

$$1 = \mathbb{E}_\theta[WU] = \mathbb{E}_\theta[(W - \theta)U] = a(\theta) \text{Var}_\theta(W).$$

Also,

$$I_n(\theta) = a(\theta)^2 \text{Var}_\theta(W) = a(\theta).$$

Consequently,

$$\text{Var}_\theta(W) = \frac{1}{I_n(\theta)}.$$

At the interior MLE,

$$0 = U(\hat{\theta}_{\text{MLE}}; X) = a(\hat{\theta}_{\text{MLE}})\{W(X) - \hat{\theta}_{\text{MLE}}\}.$$

Hence $\hat{\theta}_{\text{MLE}} = W$. It is unbiased and attains the CRLB at every parameter value. Every other unbiased estimator with finite variance has variance at least this large, so $\hat{\theta}_{\text{MLE}}$ is the UMVUE.

The common-event assumption avoids substituting a random parameter into an identity established only separately at each fixed parameter. A useful way to obtain such an event is to establish the identity on a countable dense subset and then extend by continuity in θ . In particular, suppose the densities have common positive support, so their null sets agree, and both sides are continuous in θ . Intersecting the probability-one events for the countable dense subset then produces the required simultaneous identity.

An alternative final condition is that $W \in \Theta$ almost surely and the score representation holds for every θ . Since the preceding calculation gives $a(\theta) = I_n(\theta) > 0$, the likelihood is strictly increasing for $\theta < W$ and strictly decreasing for $\theta > W$. Thus W is itself the unique global maximum on the parameter interval.

For example, if $X_i \sim N(\theta, \sigma^2)$ with known σ^2 , then

$$U(\theta; X) = \frac{n}{\sigma^2}(\bar{X} - \theta),$$

and $\hat{\theta}_{\text{MLE}} = \bar{X}$ is unbiased, efficient, and UMVUE.

Problem 3: EM with imperfect blood typing

The parameter space is the closed probability simplex

$$\Delta = \{(a, b, o) : a, b, o \geq 0, a + b + o = 1\}.$$

For estimation we allow the boundary, although the reported EM runs start in the interior. There are n independent individuals, with latent genotypes $G_1, \dots, G_n$ and observed reports $R_1, \dots, R_n$. The joint model for one individual is

$$\mathbb{P}_p(G_i = j, R_i = r) = g_j(p)C_{r,t(j)}.$$

The laboratory error probability is known, rather than estimated jointly with the allele frequencies.

1. Observed likelihood and EM derivation

Write $p = (a, b, o) = (p_A, p_B, p_O)$, with $a + b + o = 1$. In the order AA, AO, BB, BO, AB, OO , the genotype probabilities are

$$g(p) = (a^2, 2ao, b^2, 2bo, 2ab, o^2).$$

Let $t(j)$ be the true type associated with genotype j ; in this order,

$$(t(1), \dots, t(6)) = (A, A, B, B, AB, O).$$

The reporting matrix has entries

$$C_{rt} = \mathbb{P}(R = r \mid T = t) = \begin{cases} 1 - \varepsilon, & r = t, \\ \varepsilon/3, & r \neq t. \end{cases}$$

The true-type probabilities are

$$q_A = a^2 + 2ao, \quad q_B = b^2 + 2bo, \quad q_{AB} = 2ab, \quad q_O = o^2.$$

Therefore the reported-type probabilities are

$$\pi_r(p) = \frac{\varepsilon}{3} + \left(1 - \frac{4\varepsilon}{3}\right) q_r(p).$$

To verify this, separate the correct-type contribution from all errors:

$$\pi_r = (1 - \varepsilon)q_r + \frac{\varepsilon}{3}(1 - q_r) = \frac{\varepsilon}{3} + \left(1 - \frac{4\varepsilon}{3}\right) q_r.$$

Let $\kappa = 1 - 4\varepsilon/3$. Here $\kappa = 13/15 > 0$ and $\pi_r \geq 1/30$ throughout Δ . The observed count vector has a multinomial distribution:

$$\mathbb{P}_p(N_A, N_B, N_{AB}, N_O) = \frac{n!}{\prod_r N_r!} \prod_r \pi_r(p)^{N_r}.$$

The counts contain all the observed information about p , since the likelihood for the individual reports depends on them only through these four counts. For reported counts N_A, N_B, N_{AB}, N_O , the observed-data log-likelihood, omitting the multinomial coefficient, is

$$\ell(p) = \sum_{r \in \{A, B, AB, O\}} N_r \log \pi_r(p).$$

Because Δ is compact and the four π_r are continuous and bounded away from zero, ℓ is continuous on Δ and has at least one global maximizer. Existence does not imply uniqueness or guarantee that EM reaches that global maximizer.

Identifiability. For known $\varepsilon \neq 3/4$, the reporting transformation is invertible on the true-type probabilities:

$$q_r = \frac{\pi_r - \varepsilon/3}{\kappa}.$$

The allele probabilities are determined by a valid true-type vector:

$$o = \sqrt{q_O}, \quad a = 1 - \sqrt{q_B + q_O}, \quad b = 1 - \sqrt{q_A + q_O}.$$

For example, $q_B + q_O = b^2 + 2bo + o^2 = (1 - a)^2$ and $1 - a \geq 0$. Thus two allele vectors with the same reported probabilities must be equal. When $\varepsilon = 3/4$, every reported type has probability $1/4$, independent of the genotype distribution, so identifiability fails completely.

E-step. The complete-data log-likelihood for individual genotype labels is

$$\ell_c(p; G, R) = \sum_{i=1}^n \log g_{G_i}(p) + \sum_{i=1}^n \log C_{R_i, t(G_i)}.$$

Only the first sum depends on p . It depends on the genotype labels through their six counts $M_j = \sum_i \mathbf{1}\{G_i = j\}$. This explains why the six genotype counts are sufficient latent quantities for the EM M-step, despite not being observed. At iteration k , Bayes' rule gives the posterior genotype probabilities

$$w_{rj}^{(k)} = \mathbb{P}(G = j \mid R = r; p^{(k)}) = \frac{C_{r, t(j)} g_j(p^{(k)})}{\sum_{h=1}^6 C_{r, t(h)} g_h(p^{(k)})}.$$

For example, for an individual reported as type A , the six unnormalized weights are

$$\begin{pmatrix} (1 - \varepsilon)a^2 \\ (1 - \varepsilon)2ao \\ (\varepsilon/3)b^2 \\ (\varepsilon/3)2bo \\ (\varepsilon/3)2ab \\ (\varepsilon/3)o^2 \end{pmatrix},$$

evaluated at $p^{(k)}$, and their sum is $\pi_A(p^{(k)})$. The conditional expected genotype counts are

$$m_j^{(k)} = \sum_r N_r w_{rj}^{(k)}.$$

Notice that $\sum_j m_j^{(k)} = n$. Equivalently, within report group r , the latent genotype counts have conditional multinomial distribution

$$(M_{r1}, \dots, M_{r6}) \mid R, p^{(k)} \sim \text{Multinomial}(N_r; w_{r1}^{(k)}, \dots, w_{r6}^{(k)}).$$

The groups are conditionally independent. Summing their conditional means gives the displayed expected counts. The aggregate vector $(M_1, \dots, M_6)$ need not itself have one multinomial distribution conditional on the reports, because different report groups have different posterior probabilities.

M-step. Apart from terms independent of p , the expected complete-data log-likelihood is

$$\begin{aligned} Q(p \mid p^{(k)}) &= (2m_{AA} + m_{AO} + m_{AB}) \log a \\ &\quad + (2m_{BB} + m_{BO} + m_{AB}) \log b \\ &\quad + (2m_{OO} + m_{AO} + m_{BO}) \log o. \end{aligned}$$

The factors of two in the heterozygote probabilities and the reporting probabilities contribute only constants in this maximization. Define the expected allele counts by the three coefficients above. Their sum is $2n$. To maximize explicitly, put

$$A_k = 2m_{AA} + m_{AO} + m_{AB}, \quad B_k = 2m_{BB} + m_{BO} + m_{AB}, \quad O_k = 2m_{OO} + m_{AO} + m_{BO}.$$

When these are positive, the Lagrangian is

$$\mathcal{L} = A_k \log a + B_k \log b + O_k \log o + \lambda(1 - a - b - o).$$

The first-order equations are $A_k/a = B_k/b = O_k/o = \lambda$. Summing the resulting allele probabilities gives $\lambda = A_k + B_k + O_k = 2n$. The objective is strictly concave on the interior, so this is the unique M-step maximum. Thus

$$\boxed{\begin{aligned} a^{(k+1)} &= \frac{2m_{AA}^{(k)} + m_{AO}^{(k)} + m_{AB}^{(k)}}{2n}, \\ b^{(k+1)} &= \frac{2m_{BB}^{(k)} + m_{BO}^{(k)} + m_{AB}^{(k)}}{2n}, \\ o^{(k+1)} &= \frac{2m_{OO}^{(k)} + m_{AO}^{(k)} + m_{BO}^{(k)}}{2n}. \end{aligned}}$$

EM alternates these two steps from an interior starting point. For $\varepsilon = 0.10$, all reporting matrix entries are positive. If $p^{(k)}$ is interior and $n > 0$, all six genotype probabilities and all six expected genotype counts are positive. The M-step therefore remains interior. If an expected allele count is zero in a boundary run, the corresponding updated frequency is zero, with the convention $0 \log 0 = 0$ in the expected complete-data objective.

Why the observed likelihood cannot decrease. For an interior current iterate and an interior candidate p , Bayes' rule implies

$$\frac{\pi_r(p)}{\pi_r(p^{(k)})} = \sum_j w_{rj}^{(k)} \frac{g_j(p)}{g_j(p^{(k)})}.$$

Jensen's inequality for the concave logarithm gives

$$\begin{aligned} \ell(p) - \ell(p^{(k)}) &= \sum_r N_r \log \left\{ \sum_j w_{rj}^{(k)} \frac{g_j(p)}{g_j(p^{(k)})} \right\} \\ &\geq \sum_r N_r \sum_j w_{rj}^{(k)} \log \frac{g_j(p)}{g_j(p^{(k)})} \\ &= Q(p \mid p^{(k)}) - Q(p^{(k)} \mid p^{(k)}). \end{aligned}$$

The M-step maximizes Q , so its right-hand side is nonnegative at $p = p^{(k+1)}$. Thus $\ell(p^{(k+1)}) \geq \ell(p^{(k)})$. Since every $\pi_r \leq 1$, $\ell \leq 0$. The nondecreasing sequence of likelihood values therefore has a finite limit. This proves convergence of the *likelihood values*; it does not by itself prove convergence of the parameter vectors or global optimality of a limiting vector.